

Software Requirements Specifications

AI Task Agent Using More Flexible Generic Tools via Model Context Protocol (MCP)

Team Members

Jadon Duff, Computer Science (jduff2024@my.fit.edu)

Jacob Ewasko, Computer Science (jewasko2024@my.fit.edu)

Ioannis Limniatis, Computer Science (ilimniatis2023@my.fit.edu)

Jacob Lebkuecher, Software Engineering (jlebkuecher2023@my.fit.edu)

Faculty Advisor

Eraldo Ribeiro, College of Engineering and Science: Department of Electrical Engineering and
Computer Science (eribeiro@fit.edu)

Client

Doug Armstrong, District Directorate Chief - Response

of

The United States Coast Guard Auxiliary

Table of Contents

1. Introduction
 - 1.1. Executive Summary
 - 1.2. Project Scope
 - 1.3. Definitions
2. Project Description
 - 2.1. Features
 - 2.2. User Roles
 - 2.3. Operational Concept
 - 2.4. Constraints
3. Requirements
 - 3.1. Functional Requirements
 - 3.1.1. Task Execution using Generic MCP Tools
 - 3.1.2. Improved AI Knowledge via User-Supplied Documentation
 - 3.1.3. Versatile Deployment with Widespread Hardware Compatibility
 - 3.2. Interface Requirements
 - 3.2.1. User Interface Requirements
 - 3.2.2. System Interface Requirements
 - 3.3. Performance Requirements
4. Summary and Conclusion
5. References

1.0 Introduction

This document provides a detailed overview of the requirements for the AI Task Agent Using More Flexible Generic Tools via Model Context Protocol (MCP) project. This document will define an executive summary of the AI Task Agent, defining the scope and the requirements of the project.

1.1 Executive Summary

This document serves as the Software Requirements Specifications for the AI Task Agent Using More Flexible Generic Tools via Model Context Protocol (MCP) project (ATA). The ATA is the proposed system for the Fall 2026 Computer Science/Software Engineering Senior Design project. The ATA is an agentic system that encompasses (a) a fine-tuned LLM, (b) a deployment environment/HELM chart, (c) a Model Context Protocol (MCP) server, (d) a virtual machine (VM), (e) AI memory management, encompassing Retrieval-Augmented Generation (RAG) and context window management, and (f) a graphical user interface (GUI). The project will design a system such that a user can interact with a GUI to an AI Agent that has full computer access.

1.2 Project Scope

Currently, MCP servers are tuned to a single domain and preprogrammed tools. The ATA will design an MCP server with tools dynamic enough to allow an AI Agent to execute most human-like actions on a VM, resolving this issue.

LLMs are also constrained by their cutoff window (memory) and tuned for general versatility. This project will provide an automated manner to tune an LLM for a specific use case. The ATA will also design a system to handle dynamic context windows and AI memory management to allow the AI Agent to work with large systems.

Finally, a multitude of systems and APIs can lead to security concerns and vulnerabilities. This project will allow for (a) siloed deployments (e.g. each subsystem separately, matching current “more vulnerable” systems), and (b) secure air-gapped deployments (all subsystems, including LLM, local).

The uniqueness of this project is that, while some of these individual subproblems have been attempted in industry; some have not been solved disparately, others have not been solved together, and the system has not been packed for dynamic deployment.

1.3 Definitions

Term	Definition
POC	The Proof-of-Concept stage of development, coming before Prototype
Prototype	The development stage at which a working prototype is delivered, which may or may not meet requirements. Comes before MVP
MVP	The Minimum Viable Product that, at minimum, meets the requirements.
PROD	The development stage at which the system exceeds requirements and is thoroughly tested.
CGAUX	The United States Coast Guard Auxiliary; the noncombat, civilian, nonmilitary subset of the US Coast Guard.

2.0 Project Description

This section describes the high-level features, user roles, operational concepts, and constraints. Viewed from the perspective of the userbase and how it will operate. The AI Task Agent is a locally deployable AI Agent that will complete given tasks operated in a sandboxed virtual MCP through generic MCP tools drawing on user-supplied documentation to use as reference for its knowledge.

2.1 Features

Feature 1 (Task Execution using Generic MCP Tools)

- Uses a properly sandboxed MCP that is given full computer control.
 - *Is the MCP properly sandboxed?*
 - *Can the agent somehow leave or access something outside of the sandbox?*
 - *How do we isolate processes and resources?*
- Uses a properly sandboxed MCP that is given full computer control.
 - *Can we balance a small LLM with complex tool selection and reasoning?*
 - *What if the LLM selected can not map the task into steps?*
- The agent performs tasks using a combination of generic hardcoded tools to operate the computer.
 - *Can we find a small, generic list of tools?*
 - *What about edge-cases or obscure goals?*
- Mapping of tasks to steps and shared tools are in the model's context window.
 - *Can the small model sufficiently handle the tools in its context window?*
 - *Can we balance sufficient tools with a large enough context window?*
- For air-gapped systems without network access, generic tools that need network access are removed via initial configuration.
 - *Will the user understand how and which tools to disable?*
 - *What if the user leaves a tool enabled, and it errors out?*

Feature 2 (Improved AI Knowledge via User-Supplied Documentation)

- An MCP tool that searches the RAG for model-determined “search keywords” utilizing two types of supporting documents.
 - *How do we communicate to the user that their documents must be reliable?*
 - *What if the LLM is not good at searching context with the tool?*
- A normal embedding model injects context into the prompt.
 - *How do we choose good parameters for how much, and which, context to pull from the vector store?*
 - *How do we provide this to the model efficiently?*

Feature 3 (Versatile Deployment with Widespread Hardware Compatibility Tools)

- Tools are selected by the user during deployment; should the user not desire a specific tool or toolset (e.g. disabling running code or Internet access), they will do so during installation wizard.
 - *Can we trust the user to complete this step correctly?*
 - *For example, a tool would error if they leave Internet-requiring tools enabled on an air-gapped system.*
- Utilize A/B software for update versioning for systems connected to the Internet.
 - *How can we implement the versioning system?*
 - *How can we make this step automatic once we push an update?*
- Systems can consolidate LLM, MCP, and client locally.
 - *How can we integrate all 3 of these systems on resource constrained systems?*
 - *What minimum system requirements are needed?*
- System can be Internet isolated (air-gapped) or connected to the Internet; dynamic installation allows for flexibility.
 - *How do we ensure our system continues to function without Internet?*
 - *What happens if it tries to access the Internet and is denied?*
- For air-gapped systems without network access, this feature is not available and manual update is needed.
 - *Would we need to dynamically test for Internet access?*
 - *How will users manually update?*

2.2 User Roles

The primary user role using the system is Developer. However, the system will be designed with use cases in mind for non-technical users and cybersecurity-minded users.

The Developer user will use the system to expand access to LLMs and, when used locally, become a no-cost, secure system for LLM interface.

The Non-Technical user role will use the system for interaction with an LLM that has access to a computer and the Internet. This user role will align with the disparate deployment technique, as general hardware cannot run LLMs.

The Cybersecurity user role will use the system air gapped for 100% local interface. This allows peace-of-mind interaction with the system on more capable hardware without depending on an external API provider that stores data. Furthermore, the Cybersecurity user role may tune the LLM for their specific use case, such as blue/red teaming, that may be limited on traditional LLM providers.

The ATA will have configurable deployment and LLM tuning. As such, this project will be shared with the US Coast Guard Auxiliary (CGAUX) and tuned to an example use case shared by the CGAUX.

2.3 Operational Concept

The ATA system will involve a user (actor) interacting with a client system. The client will consist of a GUI. This GUI can be a local web interface or an application, but must run locally.

This client will interact with the LLM's memory manager, the LLM, and the MCP server. The memory management will consist of RAG (and an embedding model) and the context window manager.

The MCP will control a virtual machine. The virtual machine will pass through a defined firewall with configurable rules. If the system is configured, it will then have access to the Internet.

The environment of the system will be hosted locally or consist of a local client and connect via HTTP API to external systems (e.g. LLM). Both will be configurable. For the CGAUX use case, the deployment will be local.

Normal Operational State: user can interact with LLM as defined in this Software Requirements Specification.

Degraded Operational State: system will be tested to ensure this does not occur locally. For degraded hardware or low memory conditions, the system will be throttled.

Emergency Operational State: when the system is compromised or the agent is taking undesired actions, an emergency shutdown command by the client will immediately disconnect all subsystems (<100 ms).

2.4 Constraints

This AI Task Agent will encounter various constraints in how it will be able to

- Hardware: The system must meet specific requirements, consisting of medium-to-end consumer hardware. It needs to be able to run on macOS and Windows host machines. The client must share the machine's memory and processing power at the same time; this potentially limits the size of the model.
- Software: Agent tools are built for Linux; the system must run on a Linux virtual machine on macOS and Windows hosts. Open-weight models that can be bundled have to have permits for local use, fine tuning, and redistribution. The MCP server and client must conform to the Model Context Protocol,

- Security: The system cannot depend on cloud AI providers or any external service; they have to be fully air gapped. Agent needs root access to the environment; all actions need to be confined to virtual machines. Network traffic can limit the points the user has approved.
- Policy (Coast Guard): the Coast Guard Auxiliary handles sensitive information and may operate where connectivity is limited or unavailable, such as aircraft or vessels. Only provided and approved documentation may be used for RAG and fine-tuning. System usage must conform with the Coast Guard and Auxiliary policies.
- Schedule: The project is being developed by a team of four senior Florida Institute of Technology students on the Senior Design timeline. The team will be limited in hardware and budget available to the project, with will scale the amount of fine-tuning that can be performed.

3.0 Requirements

The following tables detail the Functional Requirements, Interface Requirements, and Performance Requirements.

The requirements are defined as a three letter project identifier, followed by a three letter requirements identifier, followed by a three number ID.

3.1 Functional Requirements

The *Functional Requirements* describe what the system shall do.

3.1.1 Task Execution using Generic MCP Tools

ID	Requirement
ATA-FTE-001	The system shall incorporate an AI Agent via Large Language Model (LLM) that takes textual input and produces textual outputs.
ATA-FTE-002	The system shall expose a Model Context Protocol (MCP) server to the AI Agent, and, via this standardized protocol, allow execution of one of more called tools should the AI Agent produce requests.
ATA-FTE-003	The MCP Server shall have full control of a virtual machine; with Full Control being defined as sudo (root) access to a common Linux distribution, and execution of any arbitrary text string in the Linux command line.
ATA-FTE-004	The MCP shall be properly sandboxed, meaning that the MCP tools exposed to the AI Agent do not leave the virtual machine except in specific situations. Specific situations means that the developer intended for that tool to egress from the Virtual Machine and the user provides a form of approval.
ATA-FTE-005	The AI Agent shall map its given task into multiple steps, which shall be logged. The AI Agent shall use “Thinking” or “Reasoning” output modes in its task mapping. These task mappings shall be fully AI generated.

ATA-FTE-006	The MCP server shall be HTTP-based or STDIO-based transport protocol for security.
ATA-FTE-007	The MCP server shall enforce some form of Authentication, which shall be enforced by rotating Bearer token authentication.
ATA-FTE-008	The MCP server shall enforce some form of Authorization, meaning that the tools shall not be executed without prior approval of the User. Note that approval from the User can mean approval during installation and does not require approval for every individual MCP tool.
ATA-FTE-009	The system shall allow the AI Agent to install software on the virtual machine, at a minimum installation via the Command Line of the corresponding virtual machine's operating system.
ATA-FTE-010	The system shall allow the AI Agent to create, read, update, move, and delete files and folders on its virtual machine.
ATA-FTE-011	The system shall allow the AI Agent to monitor running processes on the virtual machine, defined as getting a text output format of the current running processes and ingesting these into the AI Agent's context window.
ATA-FTE-012	The system shall allow the AI Agent to schedule programs to run later, defined as execution at a specific time, assuming the system is running at that time.,
ATA-FTE-013	The system shall allow the AI Agent to execute arbitrary programs on its virtual machine, defined as the AI Agent supplying either (1) raw code or (2) the path and name of an executable file.
ATA-FTE-014	The system shall allow the AI Agent to transfer files outside of the virtual machine, such that the user can access these files.
ATA-FTE-015	The system shall allow the AI Agent to get a screenshot of the current running operating system (of its virtual machine).
ATA-FTE-016	The system shall allow the AI Agent to interact with active windows, defined as the AI Agent can create, read, update, move, and delete active windows on the screen, via simulated keyboard interaction.
ATA-FTE-017	The system shall allow the AI Agent to output simulated keyboard commands, defined as minimally the keys of a 60% keyboard.
ATA-FTE-018	The system shall allow the AI Agent to ingest imagery or video or video of the current screen (of the virtual machine), defined as instantaneous during tool execution for imagery or instantly to an agent-supplied numerical value of seconds to record video or audio. Instantly or instantaneously is defined as a period of at maximum one second.
ATA-FTE-019	If applications display a system notification, these will be queued and appended to the next prompt to the AI Agent automatically.
ATA-FTE-020	The system shall allow the AI Agent to write, read, and execute code, defined in an MCP tool exposed to the AI Agent.
ATA-FTE-021	During execution of code, should the program raise an error, this error shall be queued and routed to the AI Agent automatically. Routed to the AI Agent means the error shall be automatically appended to the next prompt to the AI Agent.
ATA-FTE-022	The system shall allow the AI Agent to add tests to a program, if supported by the programming language the AI Agent is using. The output of these

	tests shall be automatically routed to the AI Agent. Routed to the AI Agent means the error shall be automatically appended to the next prompt to the AI Agent.
ATA-FTE-023	The system shall allow the AI Agent to interface with Git. Interface with Git means allow the AI to enable Git in a project folder, if not enabled, and execute Git commands.
ATA-FTE-024	The system shall allow the AI Agent to install more programming languages if not present.
ATA-FTE-025	The system shall allow the AI Agent to interact with an Internet browser. Interact with an Internet browser means the AI Agent is able to search text queries, ingest pages (URLs), fill out forms, download and upload files, and authenticate when permitted.
ATA-FTE-026	The system shall allow the AI Agent to get the raw HTML response from a page, if desired by the Agent. Desired by the Agent means the AI Agent requests this in an MCP tool.
ATA-FTE-027	The system shall provide the Accessibility Tree of a webpage, or similar, to the AI Agent. More specifically for “or similar”, means that the webpage will be preprocessed for the AI Agent. Preprocessed means that the webpage raw HTML will be minimally reduced in size or converted to an alternative format that is easier or more efficient for the AI Agent to read.
ATA-FTE-028	The system shall allow the AI Agent to execute REST queries and get a response.
ATA-FTE-029	The system shall allow the AI Agent to execute HTTP queries and get a response.
ATA-FTE-030	The system shall allow the AI Agent to read emails in an email repository and send emails.
ATA-FTE-031	The system shall expose the date, the current time, the user’s name, and any other context deemed to be useful by the developers, to the AI Agent, at minimum during the initial prompt.
ATA-FTE-032	The system shall allow the contents of a PDF to be injected into the AI Agent.
ATA-FTE-033	The AI Agent shall support vision inputs, defined as the LLM architecture supporting at minimum JPEG images of 1000x1000 pixel size to be input.
ATA-FTE-034	The AI Agent shall support audio, either through native LLM architecture OR transcription of an audio file.
ATA-FTE-035	The system shall support ingestion of tabular data, defined as a the AI Agent reading a minimum of a 10x10 table of integers.
ATA-FTE-036	The system shall have an egress point of the VM for files, that allow files to be exposed to the user above. The system shall ensure these files cannot be executed by this system on the host machine.
ATA-FTE-037	The system shall track computer performance metrics and report to the user on demand. Computer performance metrics is defined as the current RAM usage, VRAM usage (if present), and Unified Memory usage (if present).
ATA-FTE-038	The system shall allow the AI Agent to compress folders into ZIP format.
ATA-FTE-039	The system shall allow the AI Agent to complete given Tasks. Given Tasks are user-provided prompts that tell the AI to use the applicable MCP server

	tools to complete the Task. See the Performance Requirements for the accuracy ratings of Task completion.
--	---

3.1.2 Improved AI Knowledge via User-Supplied Documentation

ID	Requirement
ATA-FAI-001	The system shall integrate Retrieval-Augmented Generation (RAG), defined as an AI model extracting textual content from a user provided text dataset.
ATA-FAI-002	The system shall support TXT, MD, PDF, and DOCX file formats at minimum for RAG, provided by the user.
ATA-FAI-003	The system shall allow the AI Agent to write to a file available for future use, which it can get the contents of via an MCP tool. This allows the AI Agent to have persistent memory outside of its context window as is defined as Dynamic Memory.
ATA-FAI-004	The AI Agent and user shall have the ability to automatically reduce the context window, with reduced context window defined as a minimum of 1 token removed from the context window.
ATA-FAI-005	The user shall have the ability to clear the AI memory at any time, with the system complying within two seconds.
ATA-FAI-006	The system shall allow the user to manually update the AI Agent's current context window or dynamic memory files.
ATA-FAI-007	The system shall automatically manage the context window of the AI Agent. Manage the context window means the system shall not fail when a high context or maximum context is reached, and instead either (1) notify the user and stop the AI Agent or (2) reduce the size of the context window automatically.
ATA-FAI-008	The system shall have some form of Fast Context available to the user and AI Agent, defined as ingestion of textual content to the AI Agent faster than the standard LLM architecture, such as grep and reporting in a project directory instead of ingesting the entire project directory.
ATA-FAI-009	The system shall allow the user to use another LLM and function the same as using the default LLM provided by the system. Function the same means to meet the requirements in the same manner as before, without penalty should a user select an LLM of lower intelligence. If the user provides an LLM of too low intelligence to complete tasks, or the user provides an LLM of too large size to run efficiently, the system is not required to notify the user of this, but the system must display a disclaimer to the user that selecting their own LLM comes with the risks as mentioned before in this Requirement.

3.1.3 Versatile Deployment with Widespread Hardware Compatibility

ID	Requirement
ATA-FDP-001	The system shall be deployable in a containerized environment, such as Docker.
ATA-FDP-002	The system shall allow the user to define which MCP server tools require manual approval.

ATA-FDP-003	The system shall come preinstalled with Python, Java, C, C++, and an Assembly language.
ATA-FDP-004	The system shall be downloadable from GitHub.
ATA-FDP-005	The system shall be installable by downloading from GitHub and installing via a wizard.
ATA-FDP-006	The system shall be installable by putting its installation wizard onto USB.
ATA-FDP-007	The system shall be installable on a computer with no Internet access.
ATA-FDP-008	The system shall allow the user to remove tools from the MCP server, hiding them from the AI Agent. This is done during installation of the system.
ATA-FDP-009	The system shall run the LLM locally on the computer system it is installed on.
ATA-FDP-010	The system shall run the MCP server locally on the computer system it is installed on.
ATA-FDP-011	The system shall run the client locally on the computer system it is installed on.
ATA-FDP-012	The system shall run and meet all applicable requirements, when installed on a computer system that does not have Internet access. Applicable requirements means all requirements that do not require Internet access.
ATA-FDP-013	For systems connected to the Internet, the system can download updates without requiring manual user installation.
ATA-FDP-014	The system shall not interface with any network, such as the Internet, without user approval. Note that user approval can be one-time approval from the user during installation.
ATA-FDP-015	The system shall run in an operational state known as the Normal Operational State more than 98% of the time, when hardware and memory permit the system to run at this state. Normal Operational State means the user can interact with the AI Task Agent as defined in these Software Requirements Specification.
ATA-FDP-016	The system shall continue run in the Degraded Operational State. This means that the system is low on memory, running on degraded hardware, or both.
ATA-FDP-017	When the system is compromised or the agent is taking undesired actions, an emergency shutdown shall transition the system to the Emergency Operational State within 5000 milliseconds. Emergency Operational State means that the processes of the AI Task Agent shall be terminated. The system shall transition to this state upon user input.
ATA-FDP-018	The user shall be able to install the system with no action taken other than downloading the files OR transferring via USB AND interacting with the installation wizard.
ATA-FDP-019	The system shall contain all applicable files if installing via USB, such as the LLM weights.
ATA-FDP-020	The system shall stop and notify the user if it is detecting a too large context window or other performance degradation.

3.2 Interface Requirements

The *Interface Requirements* describe the requirements to communicate with the user and other systems.

3.2.1 User Interface Requirements

ID	Requirement
ATA-IUI-001	The system shall allow a user to send textual data to the AI Agent thorough the GUI.
ATA-IUI-002	The system shall not log analytics or metrics data without explicit user approval. Note that approval from the User can mean approval during installation and does not require approval for every analytics or metrics log.
ATA-IUI-003	The user shall interact with the AI Agent via a Graphical User Interface (GUI) and shall submit tasks with a prompt input button.
ATA-IUI-004	The system shall accept media such as files, images, and video from the User during prompting for task completion.
ATA-IUI-005	The system shall provide a user interface for uploading documentation for the RAG subsystem.
ATA-IUI-006	A User Interface shall be provided for other requirements regarding user interaction, such as emergency shutoff.
ATA-IUI-007	The system shall expose model hyperparameters such as temperature to the user, and be selectable via a GUI.
ATA-IUI-008	The GUI shall display past and active running tasks from the AI Agent.
ATA-IUI-009	The GUI shall display past and active prompt/chat windows with the AI Agent.
ATA-IUI-010	The GUI shall allow the user to perform creation, reading, updating, deletion, and exporting of AI Agent task interactions (e.g. prompting).
ATA-IUI-011	The system shall expose an installation wizard to the user for installation of the software system.
ATA-IUI-012	The user shall be able to delete the software completely from the user's computer via a single UI element and a confirmation.
ATA-IUI-013	The user shall have the ability to revoke given approvals at any time, unless the tasks requiring approvals have already occurred for AI Agent tasks.
ATA-IUI-014	The system shall require a single UI interaction to designate an air-gapped system, disabling Internet and other networked/communication abilities.
ATA-IUI-015	The system shall have a conversion UI element exposed to the user that allows the system to be converted to an air-gapped system. The system shall NOT allow conversion from air-gapped to networked and instead would require reinstallation of the system.
ATA-IUI-016	The system shall allow the user to upload a custom GGUF file of an LLM. Allow means provide a means of uploading or pointing to a file location of the GGUF.

3.2.2 System Interface Requirements

Per IEEE the 3.2.2 System Interface Requirements detail communications between the AI Task Agent system and external systems and do not include subsystem-to-subsystem. For subsystem communications see the design document, for subsystem requirements see the other requirements sections.

ID	Requirement
ATA-ISI-001	The system shall use the host's Internet connection to transmit and receive data to other networked systems if desired by the user. Details of what is transmitted/received via the Internet is detailed in the functional requirements.
ATA-ISI-002	The system shall query GitHub upon user approval for automatic updates. Query means check the GitHub releases API for a newer designated release of the AI Task Agent system. If so, the user shall be notified and if approved shall download the new update. This requirement is only for Internet-connected versions of the system.
ATA-ISI-003	The system shall integrate with an email or other messaging service. Integrate with means allow the AI to use at a minimum the minimum required features of such messaging service to transmit and receive messages.
ATA-ISI-004	The system shall provide a single ingress and egress point with the host and virtual machine. This single ingress and egress point shall be a directory shared by both the host machine and the virtual machine.
ATA-ISI-005	The system shall download applicable external subsystems if they are too large to commit to the Git repo of this project, during installation. For example, the LLM weights would be too large to upload to GitHub, so during installation the system would download the file. This is not required during USB installation, as the files would be located on the USB.
ATA-ISI-006	The system shall pin library versions if installing via an external package provider, such as PIP.

3.3 Performance Requirements

The *Performance Requirements* describe the time, space, and accuracy requirements of the system.

ID	Requirement
ATA-PFR-001	The system shall meet all requirements on a hardware machine consisting of a minimum of 18-core CPU, 20-core GPU, 307 GB/s memory bandwidth, 64 GB unified memory or RAM+VRAM equivalent, 1 TB storage, 100-watt-hour battery, visual display, and USB port. The system may work on lower hardware specifications but must meet requirements on these specifications. These specifications mean the system is intended to work on a medium-to-high-end consumer hardware.
ATA-PFR-002	The system shall function on a host machine running MacOS. Function means it meets all requirements.

ATA-PFR-003	The system shall function on a host machine running Windows. Function means it meets all requirements.
ATA-PFR-004	The system shall be of a maximum size of 80 GB, including the LLM weights.
ATA-PFR-005	The system shall include an LLM of 10-40 billion parameters.
ATA-PFR-006	The AI Agent shall produce tokens at a minimum average rate of 7 tokens per second, per each of the Task examples in the Task-Based Performance Requirements.
ATA-PFR-007	The AI Agent shall complete each Task-Based Performance Requirements within a timeframe of 20 minutes. Meaning, 20 minutes per requirement in the Task-Based Performance Requirements.
ATA-PFR-008	A User shall download the system from the Internet in a maximum of 10 minutes for initializing the download and 90 minutes for full download (including LLM weights). This requirement may be attempted multiple times if external systems hosting dependencies such as the LLM weights are down/slow.
ATA-PFR-009	The software shall start after being closed within a period of 120 seconds, not including the time to load the LLM weights into memory.
ATA-PFR-010	A User shall submit a Task to the AI Agent within 15 seconds, with the AI Agent starting on the Task within these 15 seconds, not counting the time to type in the prompt.
ATA-PFR-011	A New User shall install the system without consulting the developers, faculty advisor, or client. A New User is defined as a user who has never used the system, or watched the system being used by another user for more than 60 seconds.
ATA-PFR-012	A New User shall open the application and submit a Task to the AI Agent within 5 minutes. A New User is defined as a user who has never used the system, or watched the system being used by another user for more than 60 seconds.
ATA-PFR-013	The system shall transfer and install via USB within 10 minutes.

3.3.1 Task-Based Performance Requirements

The following requirements are part of the Performance Requirements and measure the accuracy of the system for completing tasks, which is its overall operational goal.

ID	Requirement
ATA-OPS-001	The AI Agent shall reason during each Task and identify any subtasks it needs to complete.
ATA-OPS-002	The AI Agent shall complete Tasks requiring a minimum of seven subtasks or steps that must be completed in sequential order to complete.
ATA-OPS-003	The AI Agent shall complete Tasks requiring a minimum of three steps and running terminal commands.
ATA-OPS-004	The AI Agent shall complete Tasks requiring a minimum of three steps and installing software.

ATA-OPS-005	The AI Agent shall complete Tasks requiring a minimum of three steps and performing create, update, read and delete operations on files.
ATA-OPS-006	The AI Agent shall complete Tasks requiring a minimum of three steps and interacting with an open application, such as Notepad.
ATA-OPS-007	The AI Agent shall complete Tasks requiring a minimum of three steps and moving the mouse and pressing the simulated keyboard.
ATA-OPS-008	The AI Agent shall complete Tasks requiring a minimum of three steps and write and execute code.
ATA-OPS-009	The AI Agent shall complete Tasks requiring a minimum of three steps and interact with imagery/video.
ATA-OPS-010	The AI Agent shall complete Tasks requiring a minimum of three steps and fix a small bug in a local repository.
ATA-OPS-011	The AI Agent shall complete Tasks requiring a minimum of three steps and create a small programming project from scratch.
ATA-OPS-012	The AI Agent shall complete Tasks requiring a minimum of three steps and navigate to a page on a browser.
ATA-OPS-013	The AI Agent shall complete Tasks requiring a minimum of three steps and search up information not present in its knowledge base.
ATA-OPS-014	The AI Agent shall complete Tasks requiring a minimum of three steps and state a fact not found in its knowledge base, instead in the provided RAG context.
ATA-OPS-015	The AI Agent shall complete Tasks requiring a minimum of three steps and fill out an online form.
ATA-OPS-016	The AI Agent shall complete Tasks requiring a minimum of three steps and submit data to a website after logging in.
ATA-OPS-017	The AI Agent shall complete Tasks requiring a minimum of three steps and interacting with an Internet based system that changes something in the physical world. For example, ordering a package from an online shopping center.
ATA-OPS-018	The AI Agent shall email its developers, the faculty advisor, the professor, and the client a summary that it has completed this requirement.
ATA-OPS-019	The AI Agent shall be given a task to remember facts for future chats, intending for the AI Agent to save the given fact in its writable memory. The AI Agent's memory shall then be cleared. The AI Agent shall then be given another Task and is expected to reason to pull from its writable memory (dynamic memory).
ATA-OPS-020	The AI Agent shall be given a prompt that exceeds its context window, and the system shall not crash.
ATA-OPS-021	The AI Agent shall complete Tasks requiring a minimum of three steps and modifying an Internet artifact that is viewable to the entire Internet.
ATA-OPS-022	The AI Agent shall complete Tasks requiring a minimum of three steps and complete a Capture the Flag (CTF) cybersecurity challenge. The CTF shall be found from a reputable cybersecurity website.
ATA-OPS-023	The AI Agent shall complete this requirement 3 times with unique Task-Based Performance Requirements each attempt. To complete, 2 of the other

	Task-Based Performance Requirements shall be chosen, and the AI will be required to complete both in the same prompt.
ATA-OPS-024	The AI Agent shall take a question from a person with a status of Basically Qualified (BQ) or Operational Auxiliarist (AUXOP) in the Coast Guard Auxiliary, and output a correct answer, given Coast Guard Auxiliary documentation.
ATA-OPS-025	The AI Agent shall complete a minimum of 4 Tasks correctly given by the Client, who is part of the Coast Guard Auxiliary.
ATA-OPS-026	The AI Agent shall perform 2 Tasks accurately in an operational environment for the Coast Guard, or nonoperational environment reporting official hours to the US Coast Guard Auxiliary. If an operational environment is unavailable OR a nonoperational environment in which the User is reporting official hours via the 7029 form, the AI Agent shall instead perform 10 Tasks accurately in a simulated operational environment. Operational environment means an environment in which the User is under official government orders, is on a boat, is in an aircraft, is performing an official Coast Guard mission. Nonoperational environment reporting official hours means the User is in a state in which they are logging hours that are reported on the US CGAUX ANSC 7029 Form.
ATA-OPS-027	The AI Agent shall complete a minimum of 4 Tasks correctly given by the Faculty Advisor.
ATA-OPS-028	The AI Agent shall complete a minimum of 4 Tasks correctly given by the Florida Institute of Technology CSE4101 CS Projects 1 professor. If the professor is unavailable, another professor may give the 4 Tasks.

4.0 Summary and Conclusion

The 131 requirements fully state what the AI Task Agent should accomplish. This consists of functional, interface, performance, and task-based performance requirements. There are 39 FTE (Functional Task Execution using Generic MCP Tools) requirements, 9 FAI (Functional Improved AI Knowledge via User-Supplied Documentation) requirements, 20 FDP (Functional Versatile Deployment with Widespread Hardware Compatibility) requirements, 16 IUI (Interface User Interface) requirements, 6 ISI (Interface System Interface) requirements, 13 PFR (Performance Requirements), and 28 OPS (Operations - Task-Based Performance) requirements.

5.0 References

830-1998 - *IEEE Recommended Practice for Software Requirements Specifications* | *IEEE Standard* | *IEEE Xplore*. (n.d.). Ieeexplore.Ieee.Org. Retrieved September 23, 2026, from <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=720574>